RISK NEUTRALITY

Learning Outcome

- describe the concept of risk neutrality in derivatives pricing

As shown in the prior lesson, an option's value is not affected by actual (real-world) probabilities of underlying price increases or decreases. This realization contributed to the growing use of options as it became easier to agree on a given option's value. As we will see in this lesson, only the expected volatility—that is, gross returns R^u and R^d introduced earlier—and not the expected return are required to price an option.

We may generalize the relationship between an option's value and that of the hedge portfolios from Equation 5 of the prior lesson using the example of a call option c_0 as follows:

$$c_0 = \frac{(\pi c_1^u + (1-\pi)c_1^d)}{(1+r)^T}. \qquad 9$$

The value of the call option today, c_0, is computed as the expected value of the option at expiration, c_1^u and c_1^d, discounted at the risk-free rate, r. This shows the derivative's value as similar to any other asset in that it equals the present value of expected future cash flows. In this case, these cash flows are weighted by assumed probabilities that are consistent with risk-neutral returns on the underlying.

The **risk-neutral probability** (π) is the computed probability used in binomial option pricing by which the discounted weighted sum of expected values of the underlying, $S_1^u = R^u S_0$ and $S_1^d = R^d S_0$, equal the current option price. Specifically, this probability is computed using the risk-free rate and assumed up gross return and down gross return of the underlying as in Equation 7.

$$\pi = \frac{1 + r - R^d}{R^u - R^d}. \qquad 10$$

More specifically, π, is the risk-neutral probability of an increase in the underlying price to $S_1^u = R^u S_0$, and (1 − π) is that of a decrease, $S_1^d = R^d S_0$.

Substituting the details from our earlier example, where $R^u S_0$= €110, $R^d S_0$= €60, S_0= €80, and X = €100, R^u = 1.375 and R^d = 0.75 and assuming an annual risk-free rate of 5%:

$$\pi = \frac{1 + r - R^d}{R^u - R^d} = \frac{1 + 0.05 - 0.75}{1.375 - 0.75} = 0.48.$$

$$c_0 = \frac{(\pi c_1^u + (1-\pi)c_1^d)}{(1+r)^T}$$

=[0.48 × Max(0, €110 − €100) + 0.52 × Max(0, €60 − €100)]/1.05

= €4.80/1.05 = €4.57, which matches our earlier result from the prior lesson.

In Equation 7, the risk-neutral probabilities are determined solely by the up and down gross returns, R^u and R^d, representing underlying asset volatility and the risk-free rate used to calculate the present value of future cash flows. This no-arbitrage derivative value established separately from investor views on risk is referred to as **risk-neutral pricing**.

The use of risk-neutral pricing goes well beyond the simple one-period binomial tree and may be applied to any model that uses future underlying asset price movements, as we will see later in the curriculum.

Revisiting Example 1 and the European call option sold by Hightest Capital, we can now explore the option price using risk-neutral pricing. Answer the following questions:

1. What is the risk-neutral probability of an up move and a down move in the one-period binomial model described in Example 1?

Hide Solution

Solution:

The risk-neutral probability of an up move, denoted π, is:

$$\pi = \frac{1 + r - R^d}{R^u - R^d} = \frac{1 + 0.05 - 0.8}{1.2 - 0.8} = 0.625.$$

The risk-neutral probability of a down move is therefore:

1 – π = 1 – 0.625 = 0.375.

2. Demonstrate how this risk-neutral probability can be used to arrive at the no-arbitrage price of the call option.

Hide Solution

Solution:

The call option price today is given by the (risk-neutral) expected value of the option payoff at maturity, discounted at the risk-free rate, *r*. In Equation 6:

$$c_0 = \frac{\left(\pi c_1^u + (1 - \pi) c_1^d\right)}{(1 + r)^T}$$

=[0.625 × Max(0, $60 − $55) + 0.375 × Max (0, $40 − $55)]/1.05

$$= \frac{\$3.125}{1.05} = \$2.98$$

which matches the value in Example 1.

3. What would be the value of a European *put* option on the same stock with the same exercise price and expiration date?

Hide Solution

Solution:

Since Equation 6 is valid for any European option, the put option value with an exercise price of $55 is:

$$p_0 = \frac{\left(\pi p_1^u + (1 - \pi) p_1^d\right)}{(1 + r)^T}$$

=[0.625 × Max(0, $55 − $60) + 0.375 × Max (0, $55 − $40)]/1.05

= $5.625/1.05 = $5.36.

Note that we have used the same risk-neutral probability π as in Questions 1 and 2, since these values are a function of the binomial model for the underlying asset for a given asset volatility and risk-free rate, not the specific option being priced.

4. Confirm that the call option price computed in question 2 and the put option price computed in question 3 both satisfy the put–call parity relationship.

Hide Solution

Solution:

European call and put options with the same exercise price and maturity date must satisfy put–call parity as defined by:

Home
Study Plan
Lessons
Flashcards
Practice
Mock Exams
Game Center
Discussions
Search

Question Set: Hedging and Risk Neutrality

1. Which of the following factors influences the value of an option price when using a binomial model?

A. The risk-free rate of return
B. The level of investors' risk aversion
C. The probability of an upward price move

Hide Solution

Solution:

The correct answer is A, the risk-free rate of return. The value of an option is determined by its risk-neutral expectation discounted at the risk-free rate. In a one-period binomial model, the risk-neutral probabilities are determined only by the risk-free rate over the life of the option and the underlying asset's volatility (as measured by the up and down gross returns, R^u and R^d). Because of the ability to construct a perfect hedge of the option using the underlying asset, an option's price is independent of investors' risk aversion and the probability of the underlying price moving up (or down).

2. If the underlying asset price in a one-period binomial model can increase by 15% or decrease by 10% over the period and the prevailing risk-free rate over the period is 4%, what is the risk-neutral probability of an asset pricing decrease?

Hide Solution

Solution:

If the underlying asset price in a one-period binomial model can increase by 15%, then the up gross return is R^u = 1.15. Similarly, if the price can decrease by 10%, then the down gross return is R^d = 0.9.

The risk-neutral probability of an upward price move is:

$$\pi = \frac{1 + r - R^d}{R^u - R^d} = \frac{1 + 0.04 - 0.9}{1.15 - 0.9} = 0.56.$$

The risk-neutral probability of a downward price move is 1 – 0.56 = 0.44.

3. Which of the following *best* describes the risk-neutral pricing interpretation of the one-period binomial option pricing formula?

A. The real-world expected payoff discounted at the risk-free rate
B. The risk-neutral expected payoff discounted at the risk-free rate
C. The risk-neutral expected payoff discounted at a risk-adjusted rate

Hide Solution

Solution:

The correct answer is B, the risk-neutral expected payoff discounted at the risk-free rate. The risk-neutral pricing interpretation of the option pricing formula states that the value of an option today is its risk-neutral expected value at maturity, discounted at the risk-free rate.

in excess of the risk-free rate by _____ at the risk-free rate, _____ the call, and _____ the underlying.

Hide Solution

Solution:

If a call option is trading at a higher price than that implied from the binomial model, investors can earn a return in excess of the risk-free rate by *borrowing* at the risk-free rate, *selling* the call, and *buying* the underlying.

A synthetic risk-free asset can be created by this strategy that earns a return higher than the risk-free rate. Selling the over-priced call will provide a higher cash inflow than is required to generate the risk-free rate of return.

5. A stock's price is currently ¥8,000. At the end of one month when its options expire, the stock price is either up by 5% or down by 15%. If the risk-free rate is –0.20% for the period, what is the value of a put option with a strike price of ¥7,950?

A. ¥333.67
B. ¥299.60
C. ¥236.93

Hide Solution

Solution:

The correct answer is B, ¥299.60. Using risk-neutral pricing, we can determine the risk-neutral probability as:

$$\pi = \frac{1 + r - R^d}{R^u - R^d} = \frac{1 - 0.002 - 0.85}{1.05 - 0.85} = 0.74.$$

The risk-neutral probability of a down move is therefore 1 – π = 1 – 0.74 = 0.26. The value of a put option with an exercise price of ¥7,950 is:

$$p_0 = \frac{\left(\pi p_1^u + (1 - \pi) p_1^d\right)}{(1 + r)^T}$$

=[0.74 × Max(0, ¥7,950 − ¥8,400) + 0.26 × Max(0, ¥7,950 − ¥6,800)]/(1 − 0.002)

=[0.26 × ¥1,150]/0.998 = ¥299.60.

OPTIONAL UNTESTED SEGMENT

UNLOCKING REAL-WORLD APPLICATIONS WITH BLOOMBERG

Bloomberg is one of the main market data platforms used globally by finance professionals throughout their workflows. In this optional section, we will explore a real-world application of the CFA Curriculum on Bloomberg.

{OVME EU} OPTION PRICER EQUITY/IR: EUROPEAN VANILLA

The Bloomberg OVME EU function allows you to value a derivative using a one-period binomial model.

> "OVME <GO> allows you to price over-the-counter derivative contracts for equity underlyings, so you can accurately quantify your market exposures for options on single-name stocks, equity indices, and funds. Using an assortment of pre-trade analytical tools, you can examine curves and cashflows and perform risk and scenario analyses for listed or OTC plain vanilla options, bond futures, listed or OTC warrants, short-term interest rate futures (STIRs), and exotic options such as variance swaps. OVME uses interpolated market-implied volatilities

Home
Study Plan
Lessons
Flashcards
Practice
Mock Exams
Game Center
Discussions
Search

trading tools that allow you to manage and execute the trades you work up in the function." -Bloomberg

To explore this function, type OVME EU into the Bloomberg command line and press Enter or <GO>. Type a security—such as Ternium Argentina SA (or TXAR AR Equity)—into the top left amber "Underlying" field, and press Enter or <GO>. Toggle the "Exercise" dropdown between "European" and "American." Ensure that the "Model" dropdown is set to "BS - continuous." The Price cell at the top left Results section calculates the no-arbitrage value of the selected option using a trinomial Black–Scholes model.

© 2023 Bloomberg L.P.

Please provide feedback on this Bloomberg feature.

Discuss Filter

Discussion

What do you want to discuss... (required)

LV

I am not sure the below is correct, what do you think?

CFA: "The risk-neutral probability (π) is the computed probability used in binomial option pricing by which the discounted weighted sum of expected values of the underlying, S1u = RuS0 and S1d = RdS0, equal the current option price"

Actually, π is such that S_0 = E [S_1]/(1+r), which means the discounted weighted sum of expected values of the underlying, S1u = RuS0 and S1d = RdS0, equal the current ~~option price~~ underlying price.

similarly, π is such that the current option price c_0 is the PV of the expected value of the option at T=1.

Created 4 months ago by Lorenzo Vella

0 replies | Last Activity: 4 months ago

Reply to this Comment

© 2025 CFA Institute. All Rights Reserved.

Home
Study Plan
Lessons
Flashcards
Practice
Mock Exams
Game Center
Discussions
Search

Home
Study Plan
Lessons
Flashcards
Practice
Mock Exams
Game Center
Discussions
Search

Home
Study Plan
Lessons
Flashcards
Practice
Mock Exams
Game Center
Discussions
Search

Home
Study Plan
Lessons
Flashcards
Practice
Mock Exams
Game Center
Discussions
Search